

Xiaoqing PAN

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Department of Epigenetic mechanisms of plant reproduction, MPI of Molecular Plant Physiology

Re: Postdoctoral Research Associate

Dear Prof. Köhler,

My name is Xiaoqing, I just finished my PhD study at Hans-Knöll-Institute in Jena, Germany. I expect to have my defense in July 2026. My PhD was on double-stranded RNA metabolism in both humans and fungi. What draws me to your lab is that your group is now investigating a new pathway leading to the formation of small RNAs from transposable elements.

Your recent Nature Plants paper caught my attention. The core of the work, long-distance mobile siRNA and their potential functions in the male germline, is the kind of question my PhD prepared me to think about. I was studying how *Aspergillus fumigatus* produces small RNAs, how dicer-like proteins process dsRNA substrates, and how the resulting 21-nt 5'-U-containing sRNAs are loaded into argonaute complexes. I analyzed sRNA-seq and mRNA-seq data to characterize the small RNA landscape and the contribution of RNAi machinery to fungal biology. After the bioinformatic prediction, I further use qPCR as well as Northern blot for the experimental validation. Through this project, I studied how to grow pathogenic fungi, RNA-seq analysis and wet-lab experiments. Work has resulted in two publications on *RNA* and *Molecular microbiology*. In parallel, I also participated in a project investigating transcriptomic responses in human PBMCs. Through characterizing transcriptomic responses after fungal stimulations with a focus on host alternative splicing and A-to-I RNA editing, we surprisingly found out that one fungus can induce RNA editing as well as disrupt host RNA splicing. Work has gained me experiences analyzing multiple RNA-seq cohorts, DEG analysis via DESeq2, alternative splicing analysis via rMATS-Turbo, RNA structure analysis via SHAPE-Map and RNA editing analysis.

My background spans both sides of what the project needs. On the wet-lab side, I have hands-on experience with sRNA cloning, custom sequencing library preparation, gene expression analysis by qPCR and Northern blot, genetic engineering of fungi, and GUS reporter assays from my master's work on rice-*Xanthomonas oryzae* interactions. My master's also gave me direct experience with plant cultivation, tissue preparation, and RNA extraction. On the computational side, I built pipelines on HPC for RNA-seq, sRNA-seq, and small RNA characterization, and I have experience analyzing sequencing data from both host and microbial fractions within the same experiment.

The bigger picture of this project is to control plant traits using sRNA silencing. I can learn plant grafting technology, germline, DNA modifications and many others. At the same time, I am seeking an international collaborative work environment, as I find integration into a vibrant team important for my development and performing high-quality science. MPI is a big institute association in Germany. I also value the interdisciplinary nature of your team. I noticed your group includes members with diverse backgrounds, and I believe that bringing my perspective from fungal genetics and host-pathogen immunology could contribute fresh insights to your projects. In the future, I want to improve my writing skills for a grant (e.g., Walter Benjamin Programme).

Thank you for your time and consideration.

Best,

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SUMMARY

RNA biologist and computational transcriptomics researcher with expertise in high-throughput sequencing, gene regulation, and RNA-mediated cellular responses. Experienced in integrating molecular biology with bioinformatics to study small RNAs, RNA interference, alternative splicing, and RNA editing across host and microbial systems. Proficient in RNA-seq analysis, reproducible computational workflows (R, Bash), and sequencing-based approaches for investigating post-transcriptional regulation.

TECHNICAL SKILLS

- **Molecular Biology:** sRNA cloning, NGS library preparation, GUS reporter assays, luciferase and GFP reporter assays, Gibson Assembly Cloning, Genetic Engineering, Northern Blot, qPCR, PCR, immunostaining, LC-MS, cultivation of microorganisms (BSL-1/2), plant cultivation and tissue preparation
- **Bioinformatics:** RNA-seq, sRNA-seq, differential expression, alternative splicing, small RNA characterization and target prediction, data visualization, HPC (Slurm), UCSC, SwissProt, KEGG, NCBI
- **Programming:** R, Python, Bash, Git, Docker, Linux
- **Language:** Chinese (native), English (fluent), German (B1)

EXPERIENCES

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Characterized RNAi and tRNA fragments processing pathways in *A. fumigatus* across developmental stages (conidia, mycelium), revealing morphotype-specific RNA regulatory mechanisms
- Analyzed differential expressed genes on mRNA-, sRNA-seq datasets, uncovering post-transcriptional regulatory networks controlling fungal development and stress response.
- Validated genes of interest (sRNA, tRNA fragments, sRNA targets) via qPCR, northern blot

Human PBMC transcriptome changes upon fungal stimulations

- Analyzed pathogen-specific transcriptomic signatures in human immune cells responding to fungal pathogens, characterized differential expressed genes, alternative splicing, A-to-I RNA editing, pathogen recognition receptor pathway activation
- Built HPC pipeline, wrote custom R scripts for multi-condition cohort analysis

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analyzed EV-associated sRNA, tRNA, and mRNA cargo from primary human neutrophils during fungal challenge

Non-coding RNA analysis of rice-*Xanthomonas oryzae* interaction

- Studied how T3SS-dependent bacterial effectors from *Xanthomonas oryzae* shape host non-coding RNA regulation; performed GUS reporter assays, plant cultivation, tissue preparation, LC-MS, qPCR, and PCR
- Engineered overexpressed and knockout mutants of circular RNA and microRNA in rice, established genetic tools that enabled investigation of ncRNA-mediated regulation

Teaching

- Microbiology and Molecular Biology, MMB005, 2022-2025. Supervised master students investigating gene regulation together with Dr. Blango and other colleagues.

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany (supervisor: Dr. Matthew Blango) 2026

Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs

MSc | Bioengineering, Chinese Academy of Sciences, China (supervisor: Dr. Kuaifei Xia) 2021

Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

PUBLICATIONS

- **Pan X***, Kelani AA*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microbiol 1-12.
- **Pan X***, Bruch A*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.
- Xia K*, **Pan X***, Chen H, Xu X, Zhang M. (2023). X00-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of OsARAB in rice. Phytopathol Res 5, 22.
- Xia K*, **Pan X***, Zeng X, Zhang M. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting OsOFP3, OsNPF2.4, and OsAGO1a, respectively. J. Plant Physiol 208, 153905.
- Kelani AA, Bruch A, Riviaccio F, Visser C, Krüger T, Weaver D, **Pan X**, Schaeuble S, Panagiotou G, Kniemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

CONFERENCES

- **2025 - Oral presentation:** German speaking Mycological Society (DMykG), Germany
- **2025 - Poster:** Gordon Research Seminar - RNA Editing, Italy
- **2024 - Oral presentation:** Jena RNA Club Symposium, Germany
- **2024 - Poster:** RNA Society Meeting, UK
- **2023 - Poster:** EMBL - The Non-coding Genome Symposium, Germany

AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023

REFEREE

- Dr. Matthew Blango (matthew.blango@leibniz-hki.de)
- Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de)